

JOOD A. AL ASWAD

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| Postdoctoral Fellow, Virginia Tech

RESEARCH INTERESTS

I am a quantitative paleobiologist who combines the fossil record, Earth system modeling, and physiology to understand how ocean life responds to environmental catastrophe, and what those patterns mean for biodiversity today. My current focus is in reconstructing pre-human marine baselines for healthy reefs from Late Pleistocene and Holocene fossils and pairing that with physiological and oceanographic data to understand what we have already lost and what remains at risk.

CURRENT POSITION

Postdoctoral Fellow: Virginia Polytechnic Institute and State University

2025 – present

Laboratory of Dr. Pedro Monarrez · Department of Geosciences

Research on climate-driven invasion biology on the US West Coast; statistical reconstruction of pre-human marine community baselines using Late Pleistocene–Holocene fossils; mentoring undergraduate and graduate students; co-developing public science engagement programming.

EDUCATION

Ph.D., Geological Sciences: Quantitative Paleobiology

2025

Stanford University · Advisor: Dr. Jonathan Payne

Dissertation: Climatic and Taxonomic Controls on Ecosystem Connectedness and Simplification in Deep Time

M.S., Geological Sciences: Stratigraphy & Petrophysics

2019

Cornell University · Advisor: Dr. Teresa Jordan

Thesis: A Stratigraphic and Petrophysical Study of In-situ Geothermal Reservoir Quality of the Cambro-Ordovician Strata in the Subsurface at Cornell University, Ithaca, New York

B.S., Earth Science: Concentration in Geology

2016

George Mason University · Advisor: Dr. Linda Hinnov

Harmonic analysis of Earth tides using geodetic borehole strainmeter data

COMPUTATIONAL & TECHNICAL TRAINING

Introduction to Earth System Modeling Workshop

2026

UC Riverside · Instructor: Dr. Andy Ridgwell

Trained in the cGENIE Earth system model of intermediate complexity for simulations of climate dynamics and biogeochemical cycles in deep time. Generated simulations and discussed and interpreted them.

Best Practices in Teaching and Mentoring Workshop

2026

Virginia Tech Center for Teaching and Learning

Course Design Institute

2025

Stanford University Center for Teaching and Learning

Designed a full course and syllabus in historical biogeography, integrating scaffolded assignments and projects to support diverse learning outcomes.

Preparing Future Professors Program

2023

Mentor: Dr. K. Allison Meezan, Foothill College · Paired with a professor for one quarter; shadowed and co-taught classes; completed coursework on pedagogy and academic career preparation.

TEACH Symposium <i>Stanford University · Workshops on evidence-based pedagogy and building inclusive classroom environments.</i>	2022
Marine Invertebrate Zoology <i>University of Washington, Friday Harbor Laboratories · Grade: 99/100</i> Intensive residential summer course covering embryology, reproduction, taxonomic classification, biology, and ecology of marine invertebrates.	2021
Field Camp <i>South Dakota School of Mines and Technology · Grade: A</i> Six-week field course in South Dakota and Wyoming covering geological mapping, stratigraphic columns, cross-sections, and technical report writing.	2015

PUBLICATIONS

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- **Al Aswad, J.**, Bazzi, M., Penn, J., Monarrez, P.M., Deutsch, C., and Payne, J. Coupling of climate change and extinction drive taxonomic homogenization in deep time. *Proceedings of the Royal Society B*, In Review.
 - **Al Aswad, J.**, Hautmann, M., Singh, P., Ferrill, N.L., Al-Ramadan, K., Lehrmann, D.J., Morsilli, M.D., Koeshidayatullah, A.I., and Payne, J.L. Bivalves from the Lower Triassic of central Saudi Arabia: Systematics and calibration of Permo-Triassic boundary position within the Khartam Member of the Khuff Formation. In Preparation.
 - **Al Aswad, J.**, Penn, J., Bazzi, M., Monarrez, P.M., Deutsch, C., and Payne, J. Physiology and climate change explain unusually high similarity across marine communities after end-Permian mass extinction. *Science Advances*, March 2025. DOI: [10.1126/sciadv.adr4199](https://doi.org/10.1126/sciadv.adr4199)
 - Bazzi, M., Lloyd, W.L., Ebersole, J., Sternes, P., **Al Aswad, J.**, and Payne, J. Extinction threatens to cause morphological and ecological homogenization in sharks. *Science Advances*, October 2025. DOI: [10.1126/sciadv.aea0278](https://doi.org/10.1126/sciadv.aea0278)
 - Payne, J.L., **Al Aswad, J.**, Deutsch, C., Monarrez, P.M., Penn, J.L., and Singh, P. Selectivity of mass extinctions: Patterns, processes, and future directions. *Cambridge Prisms: Extinction*, April 2023. DOI: [10.1017/ext.2023.10](https://doi.org/10.1017/ext.2023.10)
 - Tester, J., Beyers, S., Gustafson, J.O., Jordan, T.E., Smith, J.D., **Al Aswad, J.A.**, et al. District geothermal heating using EGS technology to meet carbon neutrality goals: A case study of Earth Source Heat for the Cornell University campus. *Proceedings World Geothermal Congress*, May 2020.
 - Gustafson, J.O., Smith, J.D., Beyers, S.M., **Al Aswad, J.A.**, Jordan, T.E., and Tester, J.W. Earth Source Heat: Feasibility of deep direct use of geothermal energy on the Cornell campus. *GRC Transactions*, Vol. 42, 2018.

GRANTS & FELLOWSHIPS

Thomas D. and Janice H. Barrow Graduate Fellowship (~\$56,000)	2024–2025
Funds for DEI Grants Program, Paleontological Society (~\$10,000)	2023
<i>Secured funding and founded the PaleoSociety DEI Grants Program to support underrepresented researchers in paleontology.</i>	
Grant for DEEP Paleontology Awards, Paleontological Society (~\$12,000)	2022
<i>Secured funding and created the Diversity, Equity, and Excellence in Paleontology (DEEP) Awards from the ground up to recognize outstanding scholars from underrepresented groups.</i>	
Graduate Fellowship, King Abdullah Scholarship Program: Stanford (~\$205,000)	2019–2022
Graduate Fellowship, King Abdullah Scholarship Program: Cornell (~\$150,000)	2017–2019
Donovan Family Scholarship, Cornell University (\$2,000)	2018
Undergraduate Student Research Grant, URSP George Mason University (~\$1,500)	2015
Full Ride Scholarship, King Abdullah Scholarship Program	2011–2016

HONORS & AWARDS

Research and Innovation Postdoctoral Fellow, Virginia Tech	2026
Winifred Goldring Award: Outstanding Paleontology PhD Student <i>Association for Women Geoscientists & Paleontological Society</i>	2025
Centennial Teaching Assistant Award, Stanford University	2025
Harriet Benson Fellowship Award for Excellent Research, Stanford University	2022
Best Scientific Video: Storytelling, Stanford BioX	2020
Certificate of Highest GPA in Program, George Mason University	2016
Outstanding Senior in Earth Science Award, George Mason University	2016

INVITED TALKS

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- Mass extinction survivors with physiological tolerance to intense climate change drive global taxonomic homogenization in deep time. To be given at Geological Society of America, Colorado, 2026.
 - Synergy of climate change and biodiversity loss drive global taxonomic homogenization in deep time. Departmental Seminar Series, George Mason University, 2026.
 - Examining the compositional similarity of marine invertebrate ecosystems after the Big Five mass extinction events. Fossil Coffee Series, UC Berkeley, 2024.
 - Taxonomic homogenization of marine ecosystems after the end-Permian mass extinction. Oxford University Biogeography Workshop, Oxford, UK, 2024.
 - Physiology of marine invertebrates explains broadened biogeography after end-Permian extinction. University of Lausanne, Switzerland, 2023.

SELECTED CONFERENCE PRESENTATIONS

** denotes student mentee*

- Trujillo, V.*, Helms, L.*, Al Aswad, J.A., Monarrez, P., and Payne, J.L. Biogeographic stability of benthic marine mollusks across the end-Cretaceous mass extinction. GSA Annual Meeting, Denver, 2022.
- Tung, K.*, Anderson, M.*, Saber, S.*, Monarrez, P.M., Al Aswad, J.A., and Payne, J.L. Comparing the latitudinal ranges of genera of Mollusca and Arthropoda before and after the end-Permian mass extinction. AGU Fall Meeting, Washington D.C., 2022.
- Al Aswad, J.A. Porosity, permeability, and paleotopographic analysis: A geological interpretation of geothermal reservoir quality at Cornell University. GSA Annual Meeting, Indianapolis, 2018.
- Al Aswad, J.A. The harmonic analysis of geophysical phenomena recorded by borehole strainmeters. OSCAR Students as Scholars Symposium, George Mason University.

TEACHING EXPERIENCE

Teaching Assistant: EPS 169/269 The Sixth Mass Extinction, Stanford University	2023 x2, 2024
Devoted one full quarter exclusively to course development with Dr. Jonathan Payne: built the syllabus from scratch, designed all activities, problem sets, and lecture materials for a new course on mass extinctions. Served as head TA for two subsequent teaching quarters, personally delivering multiple standalone lectures each term. Involved in all three iterations of the course from inception through delivery.	
Teaching Assistant: EPS 136/236 Macroevolution, Stanford University	Spring 2024
Supported lecture and discussion sections for upper-division macroevolution course covering speciation, phylogenetics, and deep-time diversity patterns.	
Teaching Assistant: GEOLSCI 4 Coevolution of Life and Earth, Stanford University	Autumn 2022

Guest Lecturer: Physical Geography (Climate Change), Foothill College

Winter 2023

Taught Week 7 on climate change; designed hands-on activities including geologic time visualization and climograph exercises.

Module Instructor: Introduction to Paleobiology, Friday Harbor Laboratories / Univ. of Washington

Summer 2021

Introduced paleobiology to undergraduate and graduate students; incorporated active-learning technology (TopHat).

STUDENT MENTORING

- Undergraduate & high school research mentor and mentor assistant, Stanford SURGE, Bio-X, and Stanford Earth Young Investigators programs (Summer 2022): Victor Trujillo, Lucy Helms, Edward Huang, Kelly Tung, McKenna Sanders, Sakeena Saber.
- Peer advisor, Bay Area Graduate Pathways in STEM, 2019–2020.
- Mentoring undergraduate and graduate students at Virginia Tech, 2025–present.

OUTREACH & COMMUNITY SERVICE

Committee Member, International Fossil Coral and Reef Society Early Career Research Symposium	2026–present
Pokémon-Inspired Evolution Museum Event: Virginia Tech Museum of Geosciences	2026–present
<i>Co-created interactive public event teaching children about evolution and biodiversity.</i>	
Creator & Committee Leader: DEEP Paleontology Awards, Paleontological Society	2022–2023
<i>Founded the Diversity, Equity, and Excellence in Paleontology (DEEP) awards to recognize underrepresented scholars.</i>	
Access, Belonging & Community Liaison, Stanford Earth	2024
Program Coordinator, Science Teaching Through Art, Stanford	2021–2022
Committee Member, Paleontological Society	2021–2025
Student Representative, Paleontological Society Council	2021–2023
Coordinator, Stanford Geological Sciences Graduate Seminar Series	2019–2022
Committee Member, North California Paleontological Conference	2020
President, AAPG Student Chapter, Cornell University	2017–2019

ACADEMIC SERVICE

Reviewer: Science	2026
Reviewer: Gondwana Research	2026
Reviewer: Science Advances	2025
Reviewer: Palaeogeography, Palaeoclimatology, Palaeoecology	2024
Session Co-Chair, Linking Biodiversity Loss to Environmental Stressors (GSA)	2025
Session Co-Chair, Quantifying Organism-Environment Interactions (GSA)	2024
Session Co-Chair, Past and Future Organism-Environment Interactions (GSA)	2023

TECHNICAL SKILLS & COMPETENCIES

Statistical modeling & data science: R/RStudio (intermediate), MATLAB and Python (beginner)
Earth system modeling: cGENIE (climate dynamics, biogeochemical cycles)
Fossil identification: marine invertebrate paleobiology (bivalves)
Field skills: paleontological and stratigraphic field work, geological mapping
Design: Adobe Creative Suite, digital art
Languages: English (fluent), Arabic (conversational)